

	Case Name: Railway Bridge (5)	Sector	Construction (civil)	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Submitted by	Mieke De Saedeleer		One of nine std. scenarios	
Date	2016	Project Control	User defined distributions	
File Name	C2016-05 Railway Bridge (5)		Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project involves the demolition of an existing railway bridge and the construction of a new one. After site preparation, the work focuses on deep foundation activities for both tracks (A and B), including pile installation, drilling, and concrete works. The superstructure is then built in phases, including arch construction, walls, and structural elements with reinforcement and concreting. Finally, the project is completed with waterproofing, backfilling, roadworks, masonry, and finishing works to make the bridge fully operational.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	34	Serial/Parallel (SP)	77%
Planned Duration (PD)	195	Activity Distribution (AD)	74%
Budget At Completion (BAC)	832497.46	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	51%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	16%	13%	1.26
CRI-rho	16%	12%	1.24
CRI-tau	9%	9%	1.45

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	N/A	N/A	N/A
CRI-rho	N/A	N/A	N/A
CRI-tau	N/A	N/A	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	78%	42%	-1.43
SI	81%	37%	-1.51
SSI	11%	14%	2.20
CRI-r	15%	14%	1.94
CRI-rho	15%	13%	1.94
CRI-tau	10%	9%	1.81

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	6%	6%
PV - SPI	7%	4%
PV - SCI	7%	3%
ED - 1	6%	5%
ED - SPI	7%	4%
ED - SCI	7%	3%
ES - 1	5%	5%
ES - SPI(t)	7%	3%
ES - SCI(t)	7%	2%

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	2%	2%
CPI	2%	1%
SPI	3%	1%
SPI(t)	3%	0%
SCI	3%	0%
SCI(t)	4%	0%
0.8 CPI + 0.2 SPI	2%	1%
0.8 CPI + 0.2 SPI(t)	2%	1%

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 10 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	-3673.63	-2.40	1	0.99	0.99	1
std dev	0	9986.10	5.25	0	0.01	0.02	0
final	0	0.00	-17	1	1.00	0.92	1

2.3.3.2. Time forecasting

PD	195	Real Duration	195	Late/early	0
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	373.10	2.47	91%	100%
PV - SPI	373.30	3.09	91%	100%
PV - SCI	373.30	3.09	91%	100%
ED - 1	374.70	5.46	92%	100%
ED - SPI	375.00	5.79	92%	100%
ED - SCI	375.00	5.79	92%	100%
ES - 1	374.40	5.25	92%	100%
ES - SPI(t)	375.00	5.27	92%	100%
ES - SCI(t)	375.00	5.27	92%	100%

2.3.3.3. Cost forecasting

BAC	832497.46	Real Cost	832497.46	Over/under budget	0
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	832497.38	0.00	0%	0%
CPI	832497.38	0.00	0%	0%
SPI	833371.83	2403.45	0%	0%
SPI(t)	833593.50	1978.04	0%	0%
SCI	833371.83	2403.45	0%	0%
SCI(t)	833593.50	1978.04	0%	0%
0.8 CPI + 0.2 SPI	832666.58	462.99	0%	0%
0.8 CPI + 0.2 SPI(t)	832713.36	389.97	0%	0%